

```
from google.colab import files
uploaded=files.upload()
```

Choose Files 2 files

historical_data.csv(text/csv) - 47516935 bytes, last modified: 6/8/2026 - 100% done
fear_greed_index (1).csv(text/csv) - 90801 bytes, last modified: 6/8/2026 - 100% done
 Saving historical_data.csv to historical_data.csv
 Saving fear_greed_index (1).csv to fear_greed_index (1).csv

```
import pandas as pd
trade=pd.read_csv("historical_data.csv")
sentiment=pd.read_csv("fear_greed_index (1).csv")
```

```
#exploring the data
print(trade.head())
print(trade.info())

print(sentiment.head())
print(sentiment.info())
```

	Closed	PnL	Transaction Hash	Order :
0	0.0	0xec09451986a1874e3a980418412fcd0201f500c95bac...	520177066:	
1	0.0	0xec09451986a1874e3a980418412fcd0201f500c95bac...	520177066:	
2	0.0	0xec09451986a1874e3a980418412fcd0201f500c95bac...	520177066:	
3	0.0	0xec09451986a1874e3a980418412fcd0201f500c95bac...	520177066:	
4	0.0	0xec09451986a1874e3a980418412fcd0201f500c95bac...	520177066:	

	Crossed	Fee	Trade ID	Timestamp
0	True	0.345404	8.950000e+14	1.730000e+12
1	True	0.005600	4.430000e+14	1.730000e+12
2	True	0.050431	6.600000e+14	1.730000e+12
3	True	0.050043	1.080000e+15	1.730000e+12
4	True	0.003055	1.050000e+15	1.730000e+12

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 211224 entries, 0 to 211223
Data columns (total 16 columns):
```

#	Column	Non-Null Count	Dtype
---	--------	----------------	-------

```

12  Crossed                211224 non-null    bool
13  Fee                    211224 non-null    float64
14  Trade ID              211224 non-null    float64
15  Timestamp              211224 non-null    float64
dtypes: bool(1), float64(8), int64(1), object(6)
memory usage: 24.4+ MB
None
      timestamp  value classification      date
0  1517463000     30             Fear  2018-02-01
1  1517549400     15  Extreme Fear  2018-02-02
2  1517635800     40             Fear  2018-02-03
3  1517722200     24  Extreme Fear  2018-02-04
4  1517808600     11  Extreme Fear  2018-02-05
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2644 entries, 0 to 2643
Data columns (total 4 columns):
#   Column                Non-Null Count  Dtype
---  -
0   timestamp              2644 non-null  int64
1   value                  2644 non-null  int64
2   classification          2644 non-null  object
3   date                   2644 non-null  object
dtypes: int64(2), object(2)
memory usage: 82.8+ KB
None

```

```

import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
plt.style.use('ggplot')
pd.set_option('display.max_columns',None)
pd.set_option('display.max_rows',100)

```

```

print("Trade shape:",trade.shape)
print("sentiment shape:",sentiment.shape)

```

```

Trade shape: (211224, 16)
sentiment shape: (2644, 4)

```

```

#missing values
print("Trade missing values:")
print(trade.isnull().sum())

print("Sentiment missing values:")
print(sentiment.isnull().sum())

```

```

Trade missing values:
Account          0
Coin             0
Execution Price  0
Size Tokens      0
Size USD         0
Side            0
Timestamp IST    0
Start Position   0
Direction        0

```

```

Closed PnL      0
Transaction Hash 0
Order ID       0
Crossed        0
Fee            0
Trade ID       0
Timestamp      0
dtype: int64
Sentiment missing values:
timestamp      0
value          0
classification 0
date           0
dtype: int64

```

```
# data preprocessing
```

```

trade['Timestamp IST']=pd.to_datetime(trade['Timestamp IST'], dayfirst=True)
trade['Date']=trade['Timestamp IST'].dt.date
sentiment['date']=pd.to_datetime(sentiment['date'])
sentiment['Date']=sentiment['date'].dt.date

```

```

merged=pd.merge(
    trade,
    sentiment[['Date','classification','value']],
    on='Date',
    how='left'
)
print(merged.shape)
merged.head()

```

```
(211224, 19)
```

	Account	Coin	Execution Price	Size Tokens	Size US
0	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9769	986.87	7872.1
1	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9800	16.00	127.6
2	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9855	144.09	1150.6
3	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9874	142.98	1142.0
4	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9894	8.73	69.7

```
merged['Win']=np.where(
    merged['Closed PnL']>0,
    1,
    0
)
```

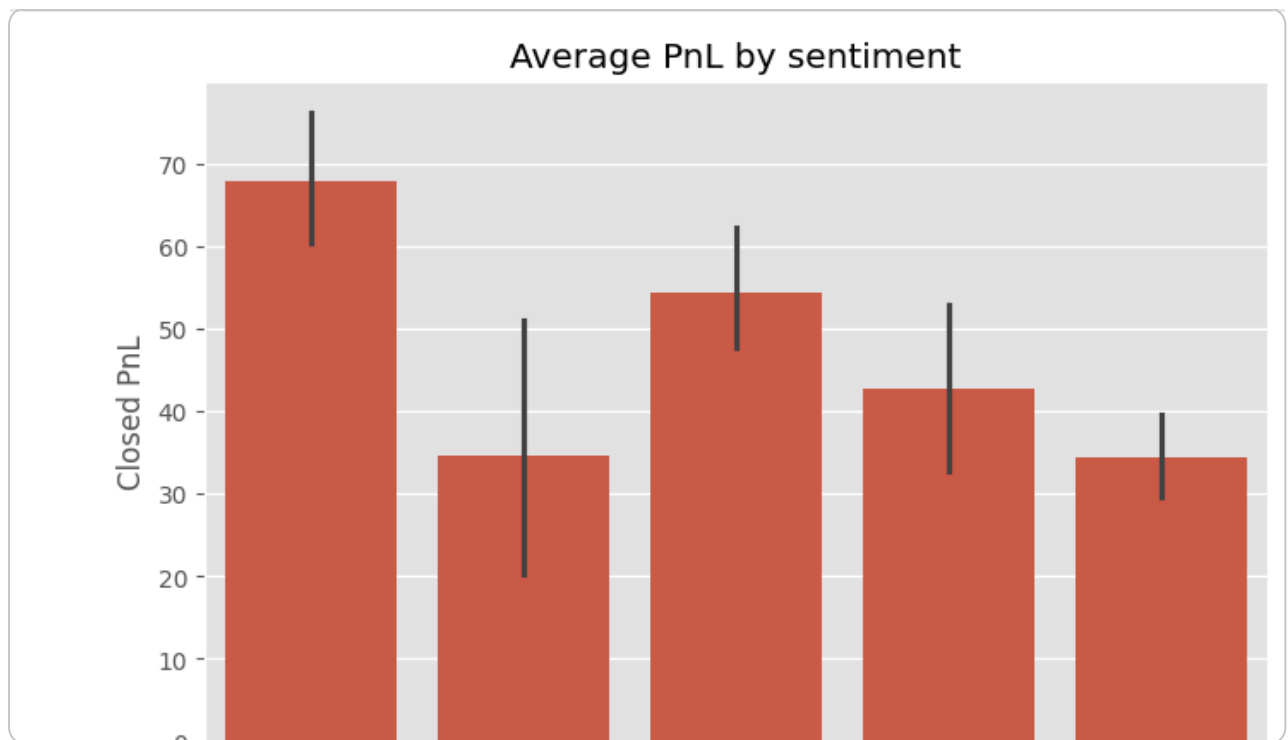
```
# calc profitabilitiy
sentiment_pnl=merged.groupby(
    'classification',
)['Closed PnL'].agg(
    ['count','mean','median','sum','std']
)
display(sentiment_pnl)
```

	count	mean	median	sum	std
classification					
Extreme Fear	21400	34.537862	0.0	7.391102e+05	1136.056091
Extreme Greed	39992	67.892861	0.0	2.715171e+06	766.828294
Fear	61837	54.290400	0.0	3.357155e+06	935.355438
Greed	50303	42.743559	0.0	2.150129e+06	1116.028390
Neutral	37686	34.307718	0.0	1.292921e+06	517.122220

Next steps:

[Generate code with sentiment_pnl](#)[New interactive sheet](#)

```
#avg pnt chart
plt.figure(figsize=(8,5))
sns.barplot(
    data=merged,
    x='classification',
    y='Closed PnL',
    estimator=np.mean
)
plt.title("Average PnL by sentiment")
plt.xticks(rotation=45)
plt.show()
```

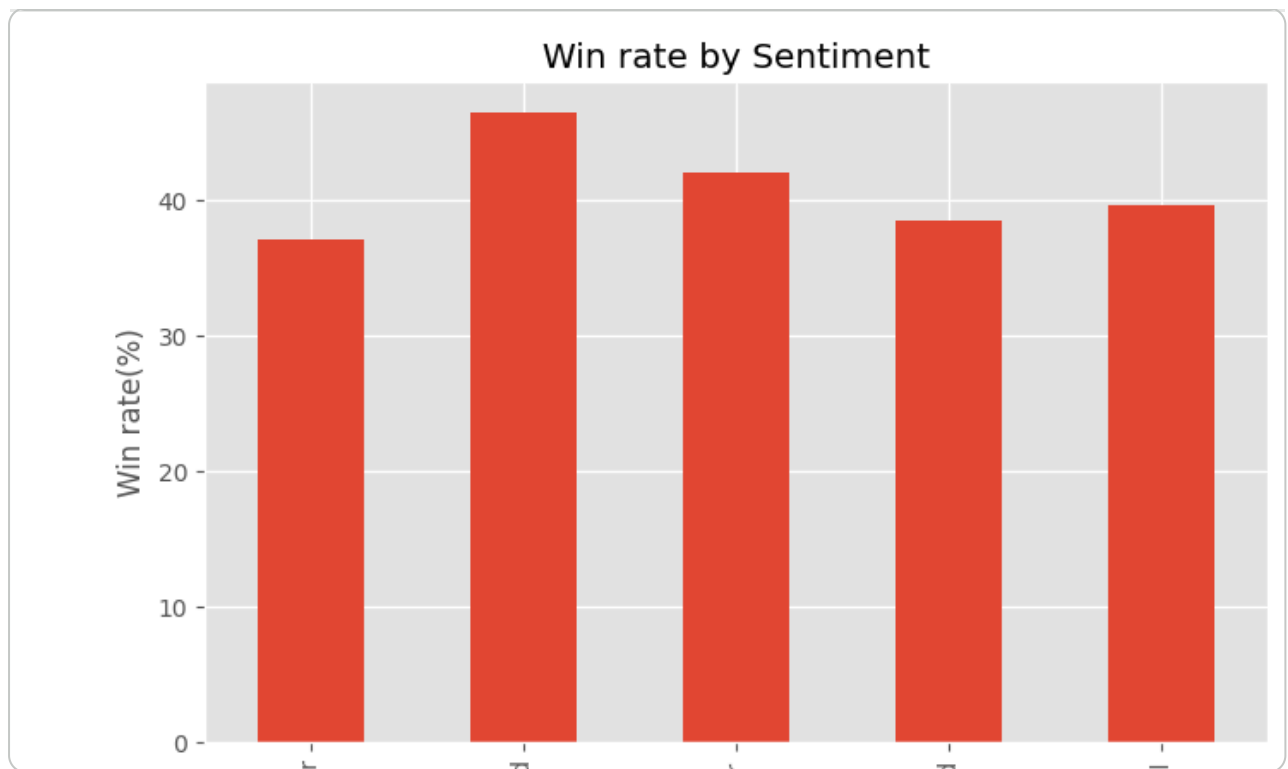


```
win_rate=(
    merged.groupby('classification')['Win']
    .mean()*100
)
display(win_rate)
```

Win	
classification	
Extreme Fear	37.060748
Extreme Greed	46.494299
Fear	42.076750
Greed	38.482794
Neutral	39.699093

dtype: float64

```
win_rate.plot(
    kind='bar',
    figsize=(8,5)
)
plt.ylabel("Win rate(%)")
plt.title("Win rate by Sentiment")
plt.show()
```

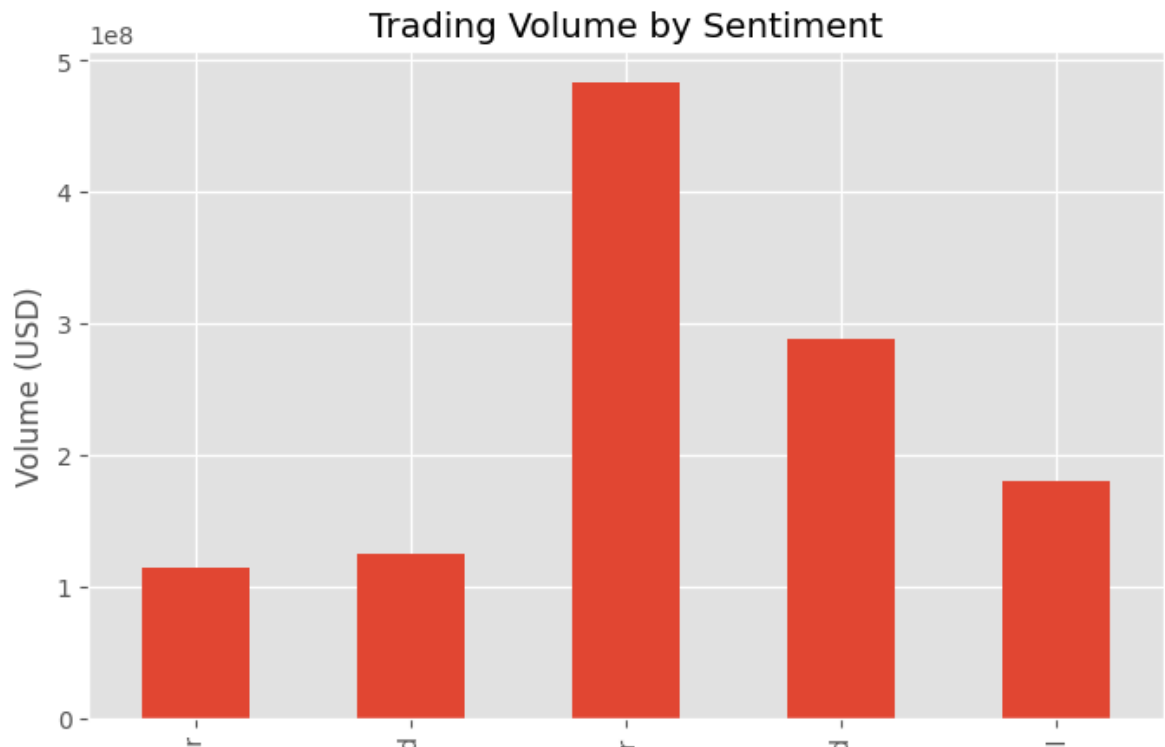


```
volume = merged.groupby(  
    'classification'  
)['Size USD'].sum()  
  
display(volume)
```

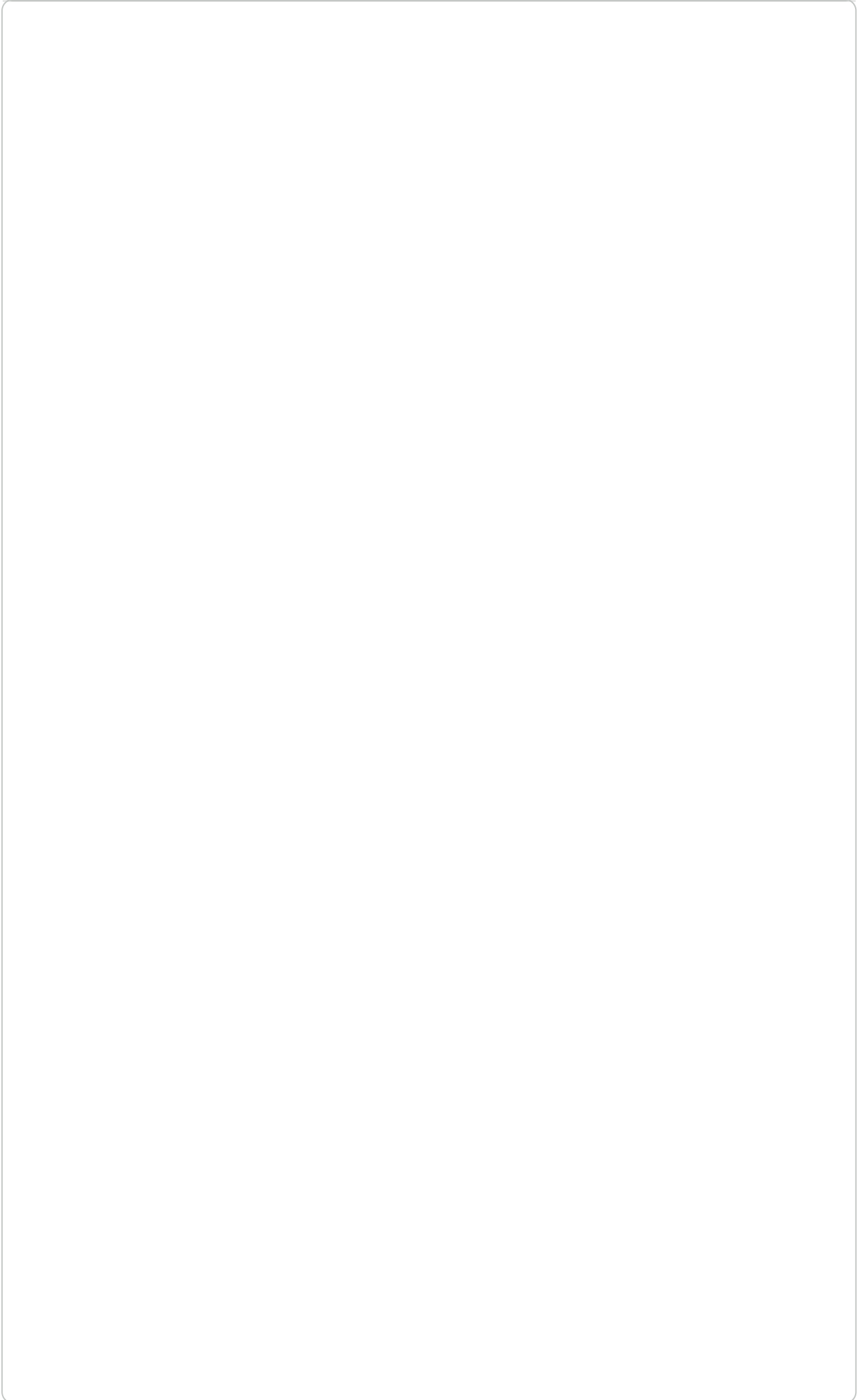
	Size USD
classification	
Extreme Fear	1.144843e+08
Extreme Greed	1.244652e+08
Fear	4.833248e+08
Greed	2.885825e+08
Neutral	1.802421e+08

dtype: float64

```
volume.plot(  
    kind='bar',  
    figsize=(8,5)  
)  
  
plt.ylabel("Volume (USD)")  
plt.title("Trading Volume by Sentiment")  
plt.show()
```



```
side_analysis = merged.groupby(  
    ['classification', 'Direction']  
)['Closed PnL'].agg(  
    ['count', 'mean', 'sum']  
)  
  
display(side_analysis)
```

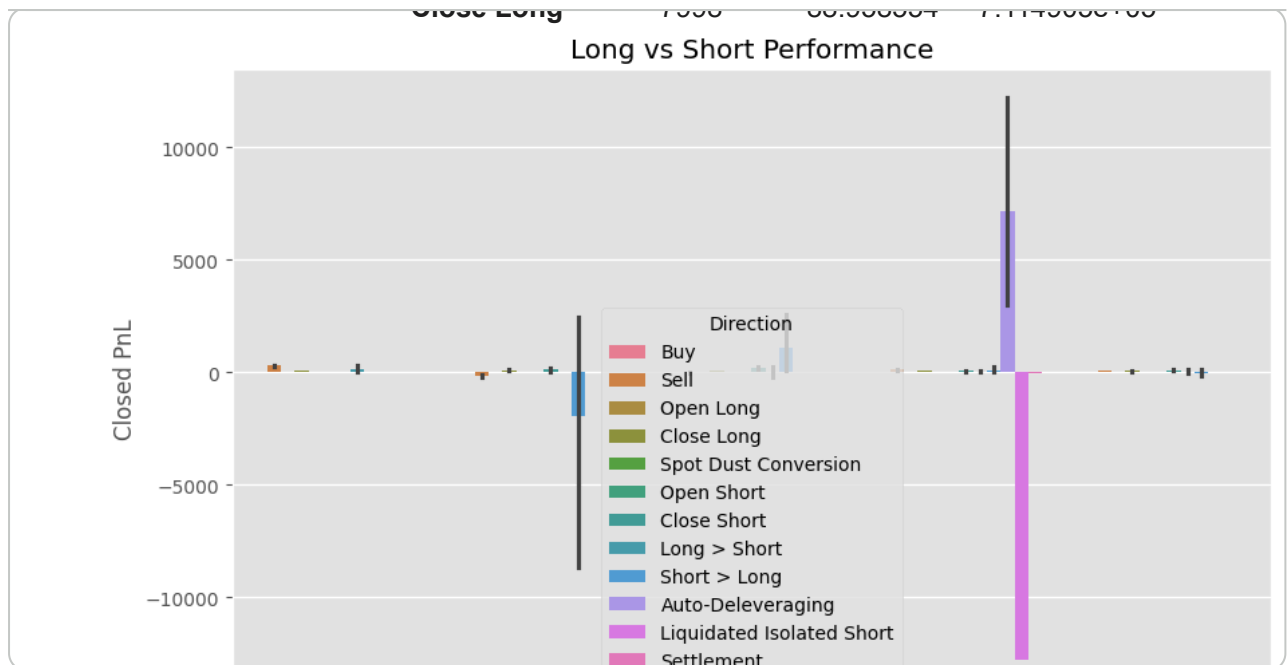


		count	mean	sum
classification	Direction			
Extreme Fear	Buy	807	0.000000	0.000000e+00
	Close Long	6241	81.197371	5.067528e+05
	Close Short	3117	123.400238	3.846385e+05
	Long > Short	5	1.331956	6.659778e+00
Next steps:	Open Long	7005	0.000000	0.000000e+00
	Open Short	3174	0.000000	0.000000e+00
	Sell	1042	-135.021730	-1.406926e+05
	Short > Long	6	-1932.516940	-1.159510e+04
	Spot Dust Conversion	3	0.000000	0.000000e+00
	Extreme Greed			
	Buy	5132	0.000000	0.000000e+00
	Close Long	7186	61.830783	4.443160e+05
	Close Short	6497	28.970490	1.882213e+05
	Long > Short	7	120.600968	8.442068e+02
	Open Long	6300	0.000000	0.000000e+00
	Open Short	7663	0.000000	0.000000e+00
	Sell	7164	290.572354	2.081660e+06
	Short > Long	11	11.771238	1.294836e+02
	Spot Dust Conversion	32	0.000000	0.000000e+00
	Fear			
	Buy	3207	0.000000	0.000000e+00

```
plt.figure(figsize=(10,6))
sns.barplot(
    data=merged,
    x='classification',
    y='Closed PnL',
    hue='Direction'
)

plt.title("Long vs Short Performance")
plt.xticks(rotation=45)
plt.show()
```

	Spot Dust Conversion	42	0.000000	0.000000e+00
Greed	Auto-Deleveraging	8	7184.807003	5.747846e+04
	Buy	4685	0.000000	0.000000e+00
	Close Long	7009	88.058554	7.114005e+05



```
fee_analysis = merged.groupby(
    'classification'
)['Fee'].mean()

display(fee_analysis)
```

	Open Long	10222	0.000000	0.000000e+00
	Fee			
classification	Open Short	6353	0.000000	0.000000e+00
Extreme Fear	1.116291	2346	89.517239	2.100074e+05
Extreme Greed	0.675902	12	-11.945100	-1.433412e+02
Fear	1.495172	10	0.000000	0.000000e+00
Greed	1.254372			
Neutral	1.044798			

dtype: float64

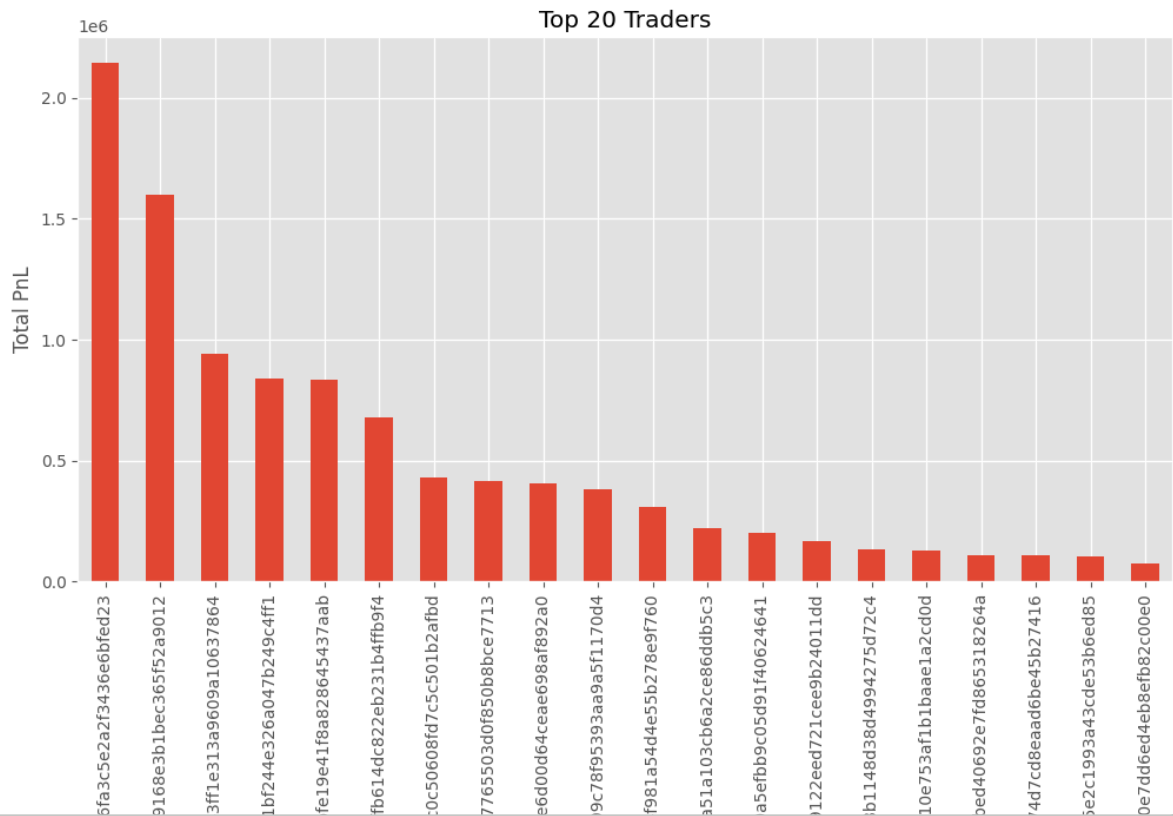
```
#top 20 traders
top_traders = (
    merged.groupby('Account')['Closed PnL']
    .sum()
    .sort_values(ascending=False)
    .head(20)
)

display(top_traders)
```

	Closed PnL
Account	
0xb1231a4a2dd02f2276fa3c5e2a2f3436e6bfed23	2.143383e+06
0x083384f897ee0f19899168e3b1bec365f52a9012	1.600230e+06
0xbaaaf6571ab7d571043ff1e313a9609a10637864	9.401638e+05
0x513b8629fe877bb581bf244e326a047b249c4ff1	8.404226e+05
0xbec1707d6b44d4d52bfe19e41f8a828645437aab	8.360806e+05
0x4acb90e786d897ecffb614dc822eb231b4ffb9f4	6.777471e+05
0x72743ae2822edd658c0c50608fd7c5c501b2afbd	4.293556e+05
0x430f09841d65beb3f27765503d0f850b8bce7713	4.165419e+05
0x72c6a4624e1dffa724e6d00d64ceae698af892a0	4.030115e+05
0x75f7eeb85dc639d5e99c78f95393aa9a5f1170d4	3.790954e+05
0x4f93fead39b70a1824f981a54d4e55b278e9f760	3.089759e+05
0xbd5fead7180a9c139fa51a103cb6a2ce86ddb5c3	2.205191e+05
0x420ab45e0bd8863569a5efbb9c05d91f40624641	1.995056e+05
0x2c229d22b100a7beb69122eed721cee9b24011dd	1.686580e+05
0x28736f43f1e871e6aa8b1148d38d4994275d72c4	1.324648e+05
0x92f17e8d81a944691c10e753af1b1baae1a2cd0d	1.265789e+05
0x6d6a4b953f202f8df5bed40692e7fd865318264a	1.087312e+05
0xa0feb3725a9335f49874d7cd8eaad6be45b27416	1.063029e+05
0x47add9a56df66b524d5e2c1993a43cde53b6ed85	1.033437e+05
0xa520ded057a32086c40e7dd6ed4eb8efb82c00e0	7.284648e+04

dtype: float64

```
plt.figure(figsize=(12,6))
top_traders.plot(kind='bar')
plt.title("Top 20 Traders")
plt.ylabel("Total PnL")
plt.show()
```



```
#top traders during fear
fear_traders = merged[
    merged['classification'].isin(
        ['Fear', 'Extreme Fear']
    )
]

fear_rank = (
    fear_traders.groupby('Account')
    ['Closed PnL']
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

display(fear_rank)
```

	Closed PnL
Account	
0x083384f897ee0f19899168e3b1bec365f52a9012	1.238143e+06
0xbaaaf6571ab7d571043ff1e313a9609a10637864	8.827904e+05
0x4acb90e786d897ecffb614dc822eb231b4ffb9f4	4.634781e+05
0x72c6a4624e1dffa724e6d00d64ceae698af892a0	3.434149e+05
0x513b8629fe877bb581bf244e326a047b249c4ff1	2.960444e+05
0xbd5fead7180a9c139fa51a103cb6a2ce86ddb5c3	2.745103e+05
0x8170715b3b381dff7062c0298972d4727a0a63b	1.640062e+05
0x420ab45e0bd8863569a5efbb9c05d91f40624641	1.091094e+05
0xb1231a4a2dd02f2276fa3c5e2a2f3436e6bfed23	1.027101e+05
0xbee1707d6b44d4d52bfe19e41f8a828645437aab	8.675289e+04

dtype: float64

```
#top traders during greed
greed_traders = merged[
    merged['classification'].isin(
        ['Greed', 'Extreme Greed']
    )
]

greed_rank = (
    greed_traders.groupby('Account')
    ['Closed PnL']
    .sum()
    .sort_values(ascending=False)
    .head(10)
)

display(greed_rank)
```

	Closed PnL
Account	
0xb1231a4a2dd02f2276fa3c5e2a2f3436e6bfed23	1.639363e+06
0xbee1707d6b44d4d52bfe19e41f8a828645437aab	7.222110e+05
0x72743ae2822edd658c0c50608fd7c5c501b2afbd	4.535953e+05
0x430f09841d65beb3f27765503d0f850b8bce7713	3.519654e+05
0x75f7eeb85dc639d5e99c78f95393aa9a5f1170d4	3.066642e+05
0x083384f897ee0f19899168e3b1bec365f52a9012	2.364369e+05
0x4f93fead39b70a1824f981a54d4e55b278e9f760	2.082412e+05
0x513b8629fe877bb581bf244e326a047b249c4ff1	1.630477e+05
0x28736f43f1e871e6aa8b1148d38d4994275d72c4	1.392026e+05
0x92f17e8d81a944691c10e753af1b1baae1a2cd0d	1.237745e+05

dtype: float64

```
#trader segmentation
trader_perf = (
    merged.groupby('Account')
    ['Closed PnL']
    .sum()
    .reset_index()
)

trader_perf['Segment'] = pd.qcut(
    trader_perf['Closed PnL'],
    q=3,
    labels=['Bottom', 'Middle', 'Top']
)

display(trader_perf.head())
```

	Account	Closed PnL	Segment
0	0x083384f897ee0f19899168e3b1bec365f52a9012	1.600230e+06	Top
1	0x23e7a7f8d14b550961925fbfdaa92f5d195ba5bd	4.788532e+04	Bottom
2	0x271b280974205ca63b716753467d5a371de622ab	-7.043619e+04	Bottom
3	0x28736f43f1e871e6aa8b1148d38d4994275d72c4	1.324648e+05	Middle
4	0x2c229d22b100a7beb69122eed721cee9b24011dd	1.686580e+05	Middle



```

segment_summary = (
    trader_perf.groupby('Segment')
    ['Closed PnL']
    .agg(['count', 'mean', 'sum'])
)

display(segment_summary)

```

/tmp/ipykernel_12685/2256706332.py:2: FutureWarning: The default of observed=trader_perf.groupby('Segment')

	count	mean	sum	
Segment				
Bottom	11	1377.819410	1.515601e+04	
Middle	10	130679.632826	1.306796e+06	
Top	11	815909.691061	8.975007e+06	

Next steps:

[Generate code with segment_summary](#)

[New interactive sheet](#)

```

#coin wise
coin_perf = (
    merged.groupby('Coin')
    ['Closed PnL']
    .sum()
    .sort_values(ascending=False)
    .head(20)
)

display(coin_perf)

```

Closed PnL

Coin	
@107	2.783913e+06
HYPE	1.948485e+06
SOL	1.639556e+06
ETH	1.319979e+06
BTC	8.680447e+05
MELANIA	3.903511e+05
ENA	2.173295e+05
SUI	1.992688e+05
ZRO	1.837778e+05
DOGE	1.475432e+05
PURR/USDC	7.526106e+04
AIXBT	7.371217e+04
BERA	7.368975e+04
USUAL	6.963194e+04
EIGEN	6.503085e+04
AVAX	4.829731e+04
ETHFI	4.392901e+04
JUP	4.290922e+04
kBONK	3.555125e+04
JELLY	3.519372e+04

dtype: float64

```
plt.figure(figsize=(12,6))
coin_perf.plot(kind='bar')
plt.title("Top Coins by Total PnL")
plt.ylabel("PnL")
plt.show()
```




```
#sentiment V/S coin
coin_sentiment = pd.pivot_table(
    merged,
    values='Closed PnL',
    index='Coin',
    columns='classification',
    aggfunc='sum'
)
display(coin_sentiment.head())
```

classification	Extreme Fear	Extreme Greed	Fear	Greed
Coin				
@1	NaN	4.956676e+02	0.000000	1.078679
@10	-0.063484	NaN	0.000000	0.000000
@100	NaN	2.762714e+01	NaN	NaN
@103	NaN	3.406334e+01	NaN	4.494691
@107	-136085.368300	1.988619e+06	-12514.843274	724342.305860

```
#CORRELATION MATRIX
numeric_cols = merged[[
    'Execution Price',
    'Size Tokens',
    'Size USD',
    'Start Position',
    'Closed PnL',
    'Fee',
```